

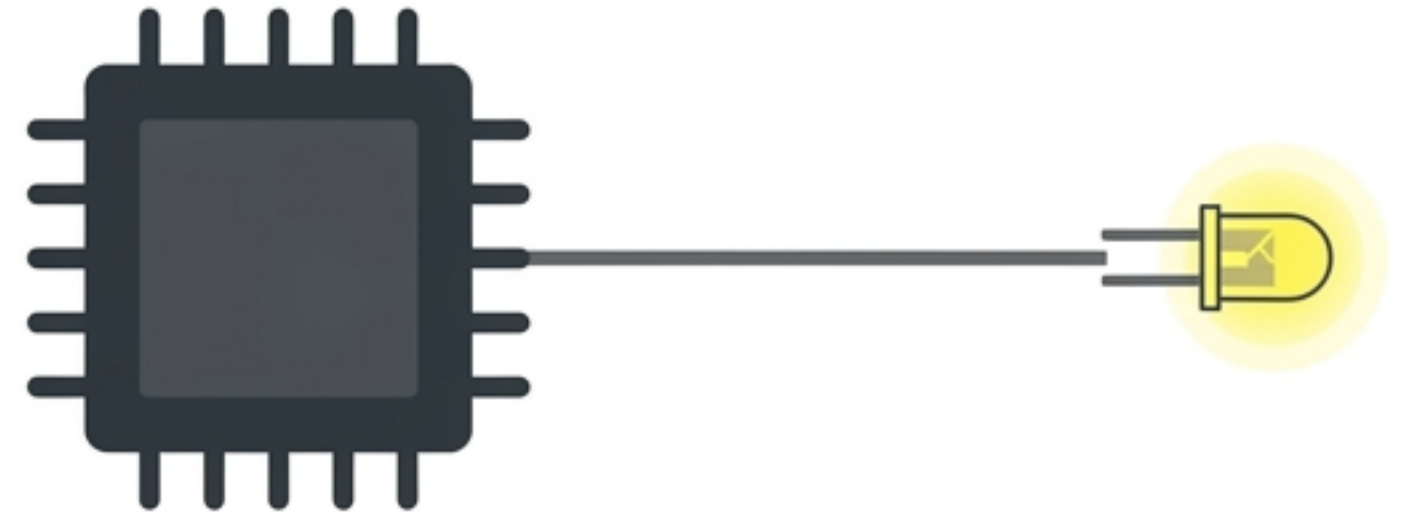
XXXXXXX

Module 3: Basic Wiring and Electronics

Course: IIT67-372 Internet of Things

Lecturer: Asst. Prof. Dr. Chanankorn Jandaeng

School of Informatics, Walailak University, Thailand

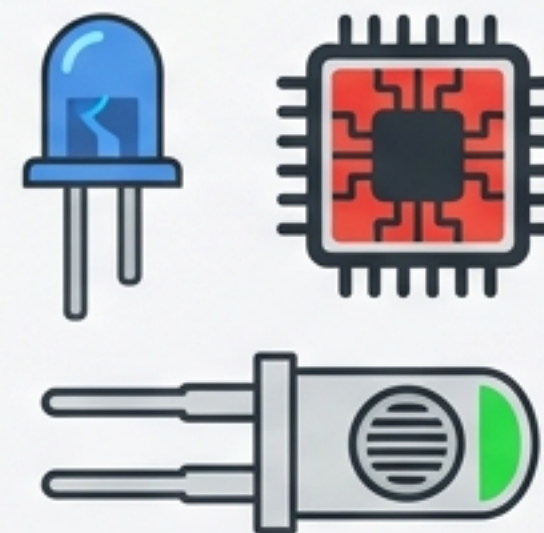


Software needs a physical body to interact with the real world



The Brain

Software & Logic
(Invisible)

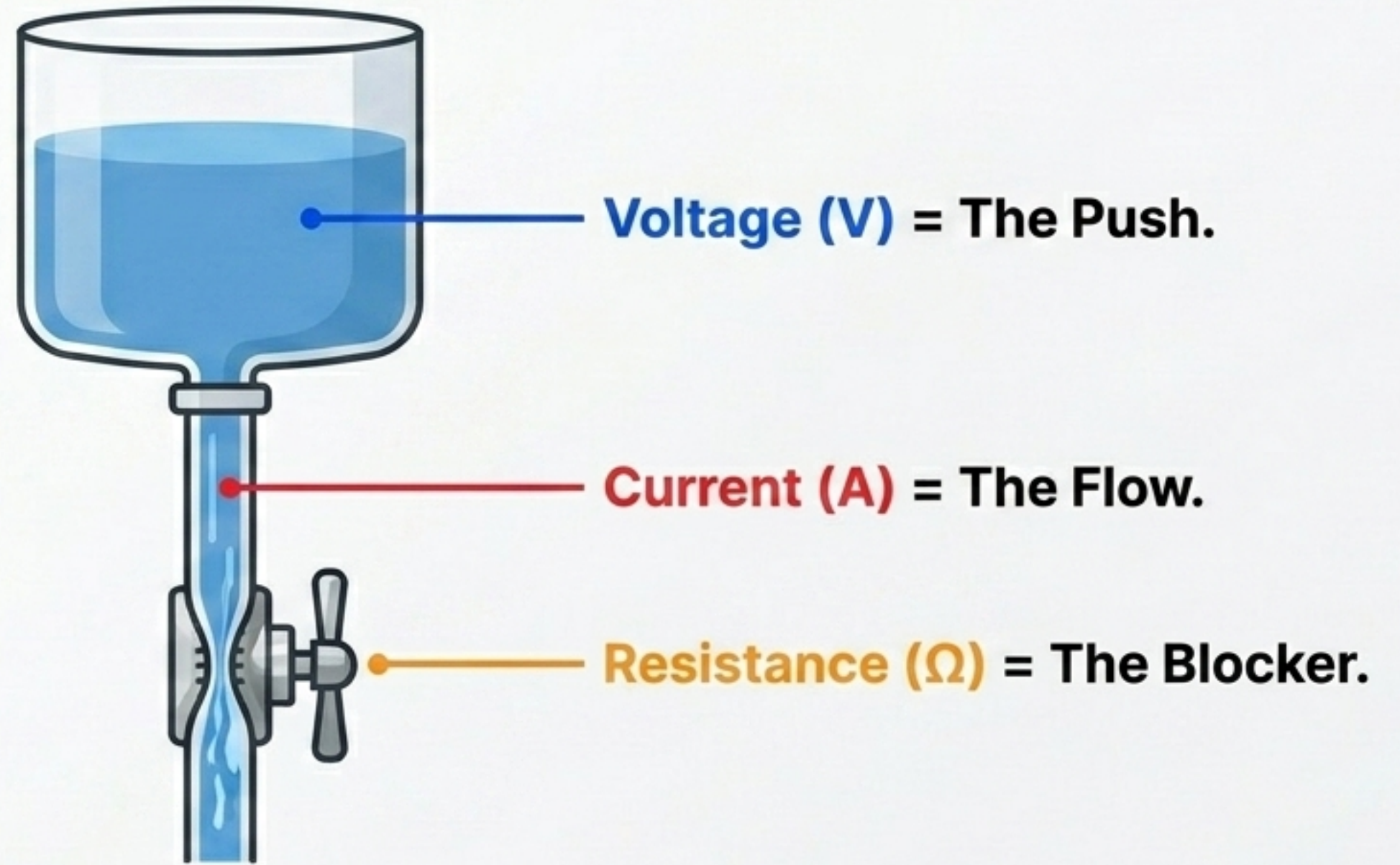


The Body

Hardware & Electronics
(Physical)

Before writing code to control the world, you must learn to wire the **physical bridge**.
This module is your foundation for safe IoT prototyping.

The forces inside the wire

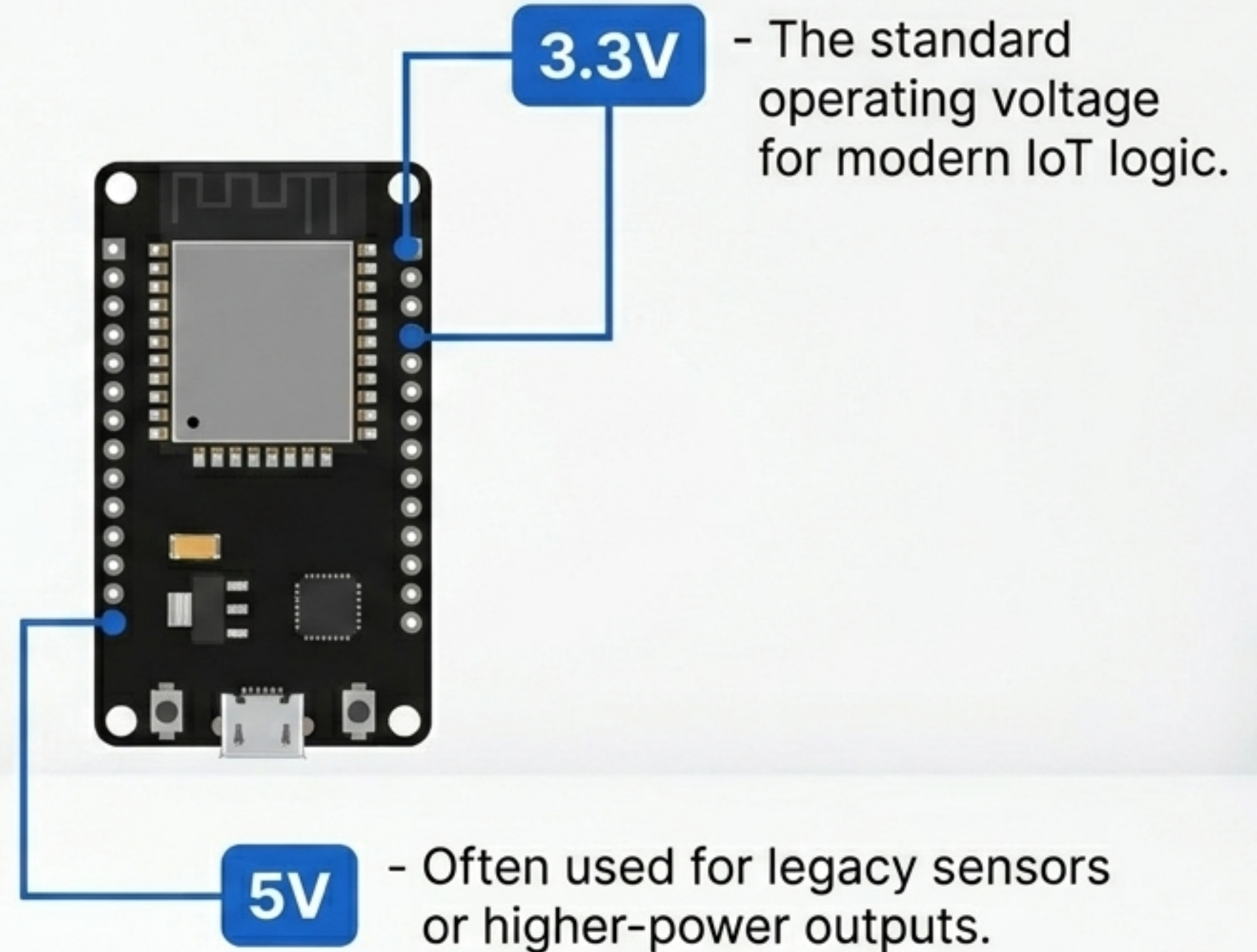


Electricity is simply the controlled flow of invisible energy. We manage the **Push**, monitor the **Flow**, and use **Blockers** to keep our components safe.

Voltage provides the push



Measured in **Volts**.
It represents the potential energy
ready to push electrons through
your circuit.



Controlling the flow to prevent damage

High Current
(Amperes)



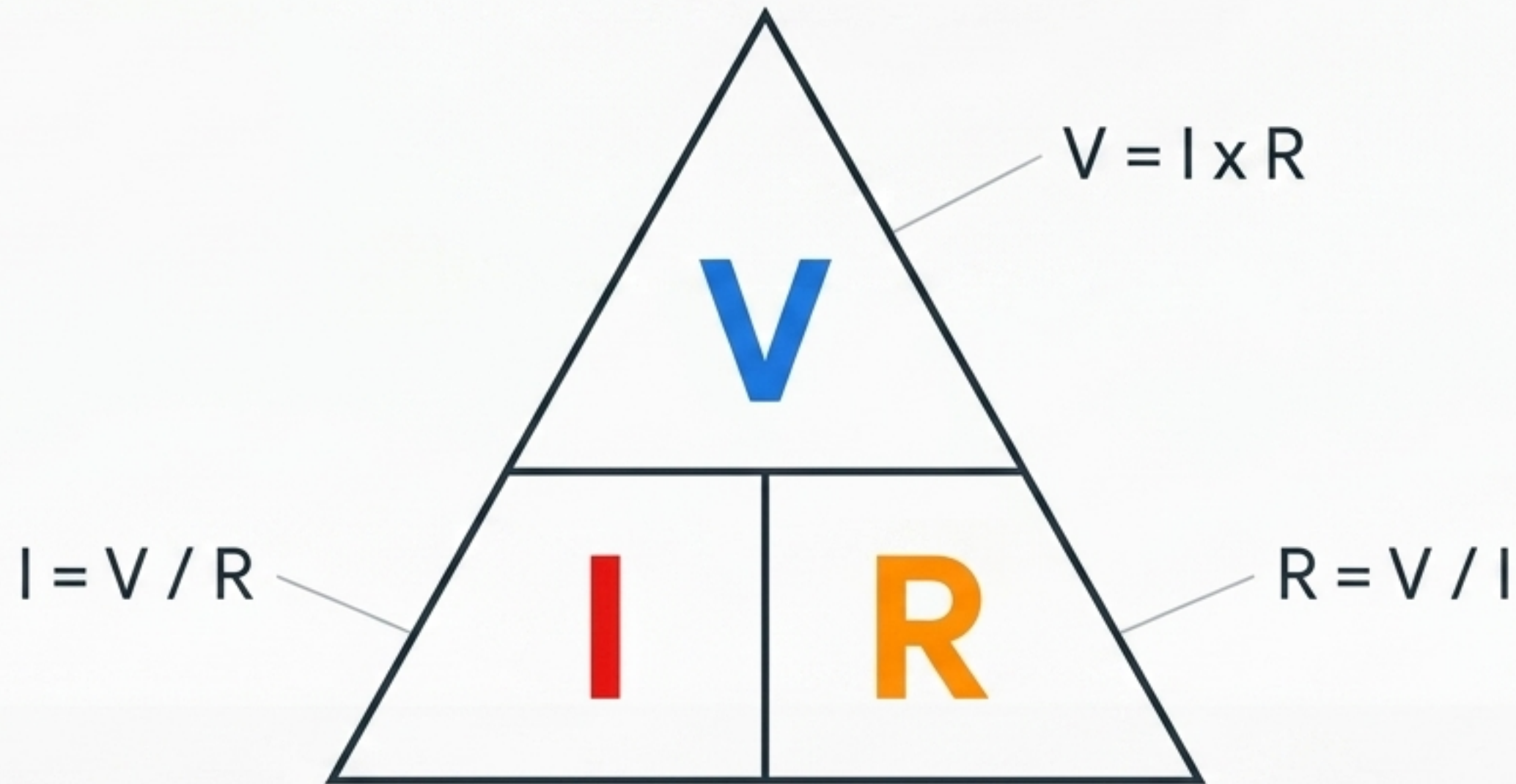
Unrestricted flow will instantly destroy sensitive IoT hardware.

Resistance
(Ohms / Ω)
applied



Resistors limit the current to safe, functional levels.

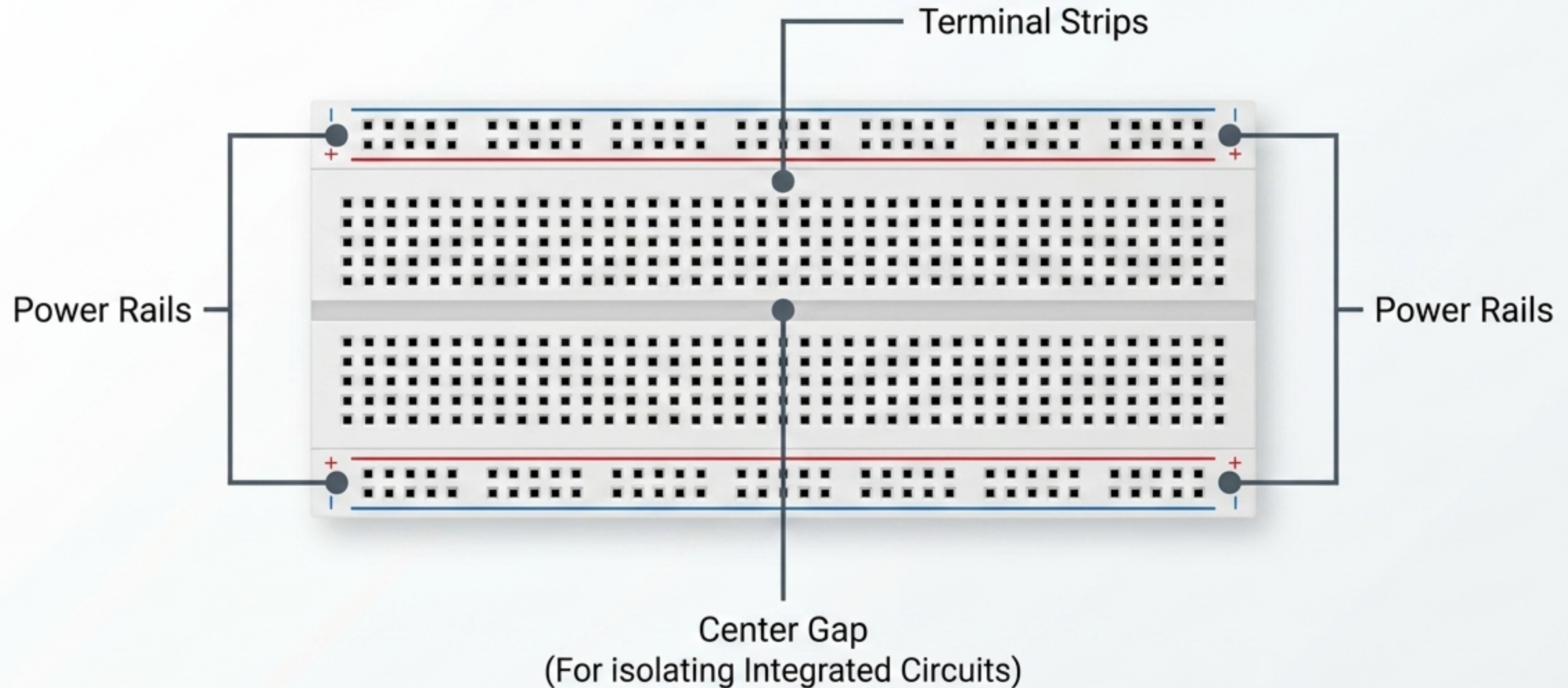
Ohm's Law governs safe circuit design



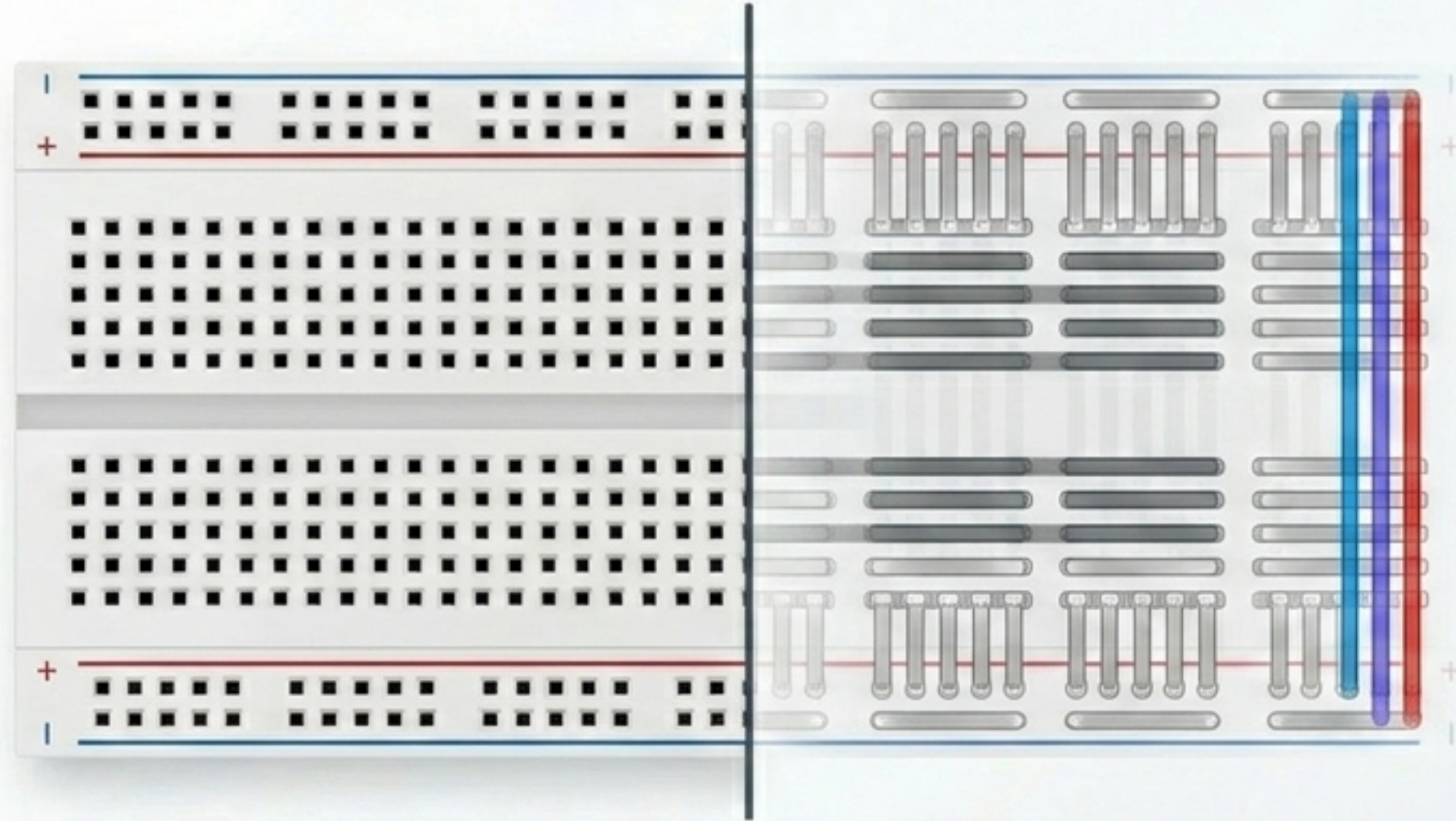
Understanding this relationship ensures you never apply too much current to your IoT sensors and outputs.

The prototyping canvas

Solderless Prototyping - Build, test, and rebuild circuits without permanent connections.

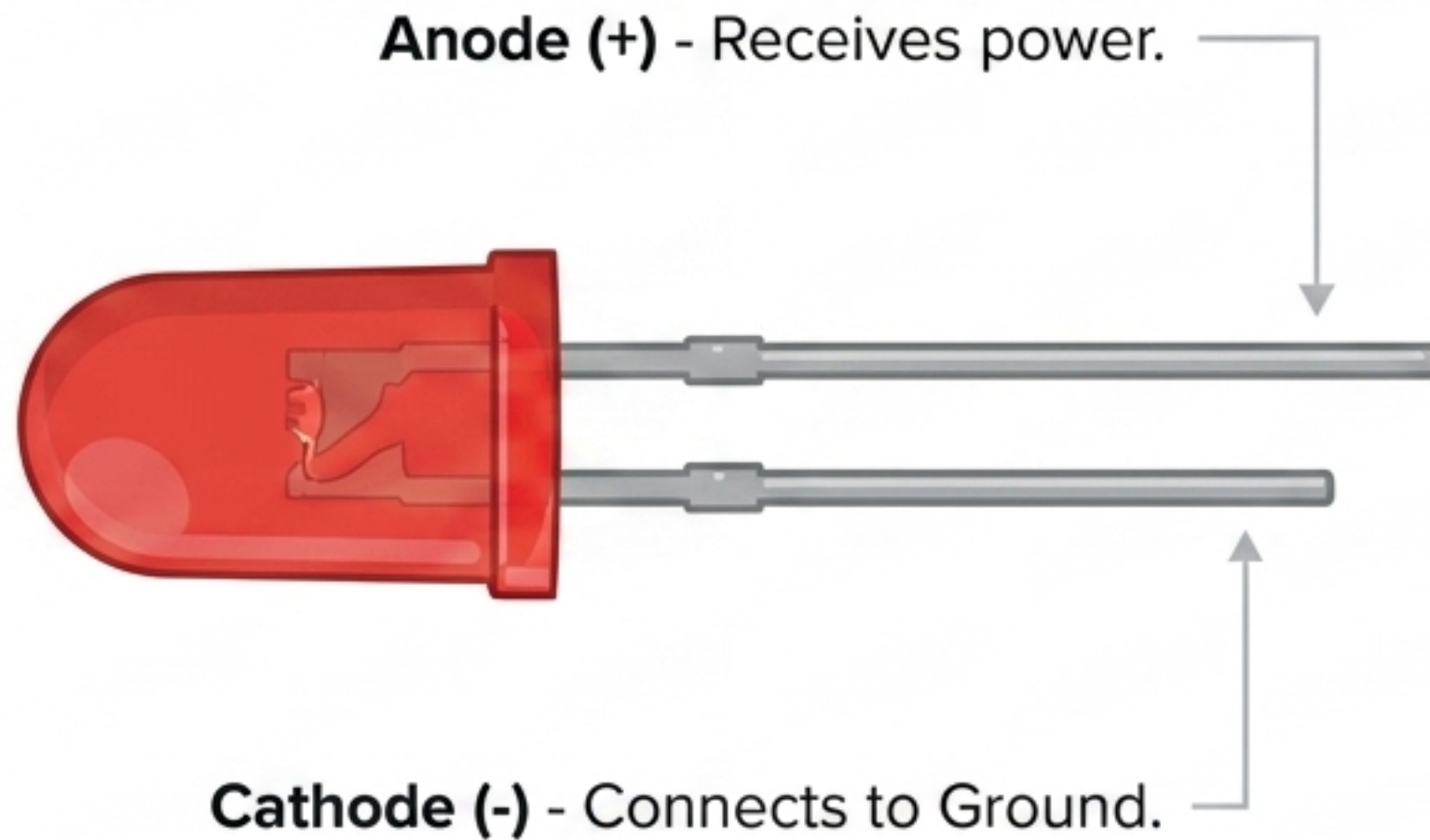


What hides beneath the plastic

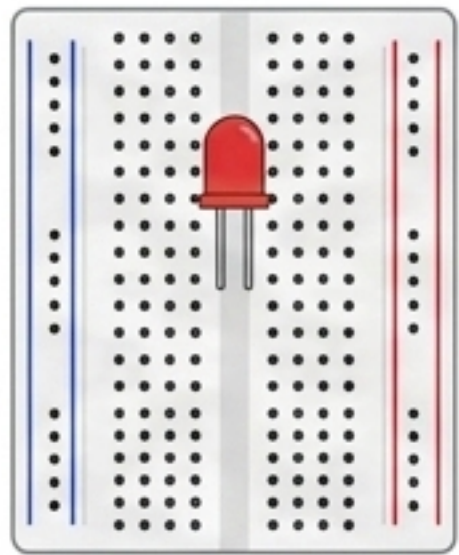


Rule of Thumb: If components share the same 5-hole row, they are electrically wired together.

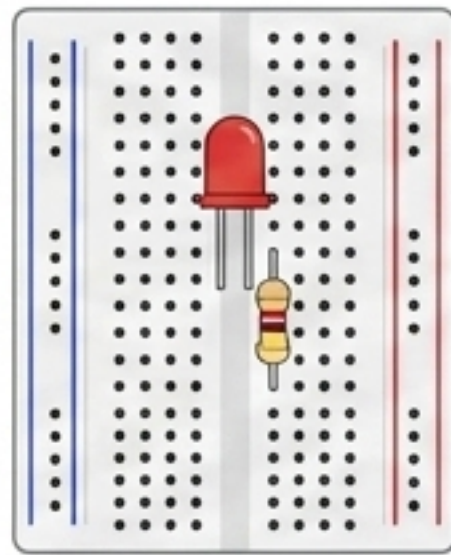
The basic output toolkit



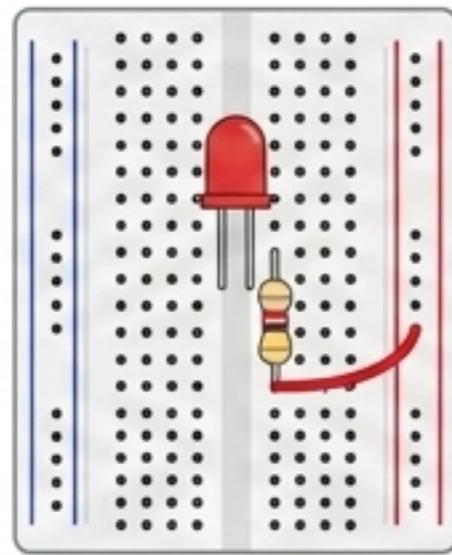
Lab Protocol: The basic LED circuit



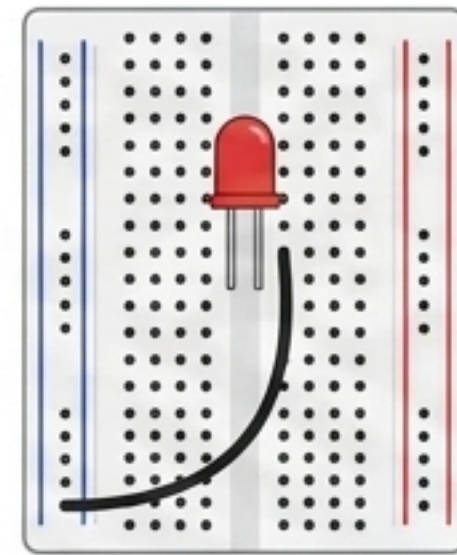
1. Insert LED



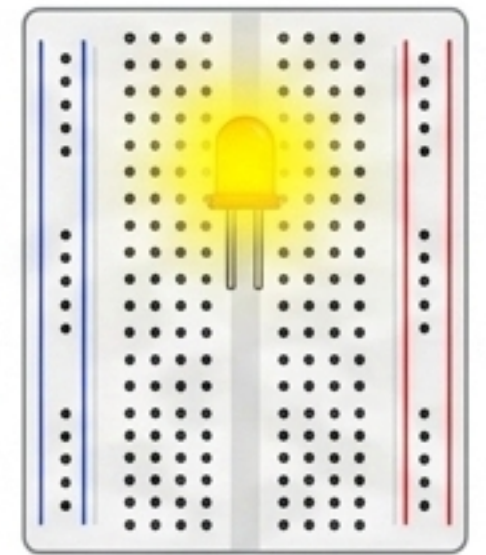
2. Add Resistor



3. Power Line
(5V)



4. Ground Line
(GND)

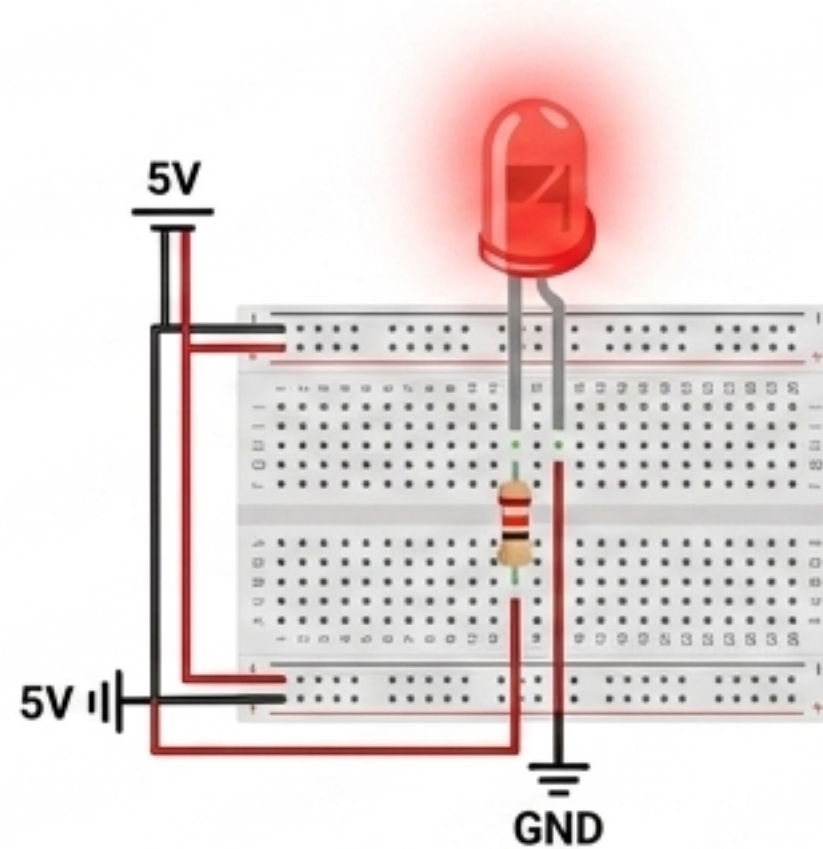


5. Activate



Objective: Learn to safely control a basic hardware output.

If you can blink an LED, you can control the world



The LED is the foundational "Output".

The logic used to trigger this light is identical to the logic used to trigger heavy industrial machinery.



Visual Indicators

(Server status, Network connection state)



Mechanical Actuators

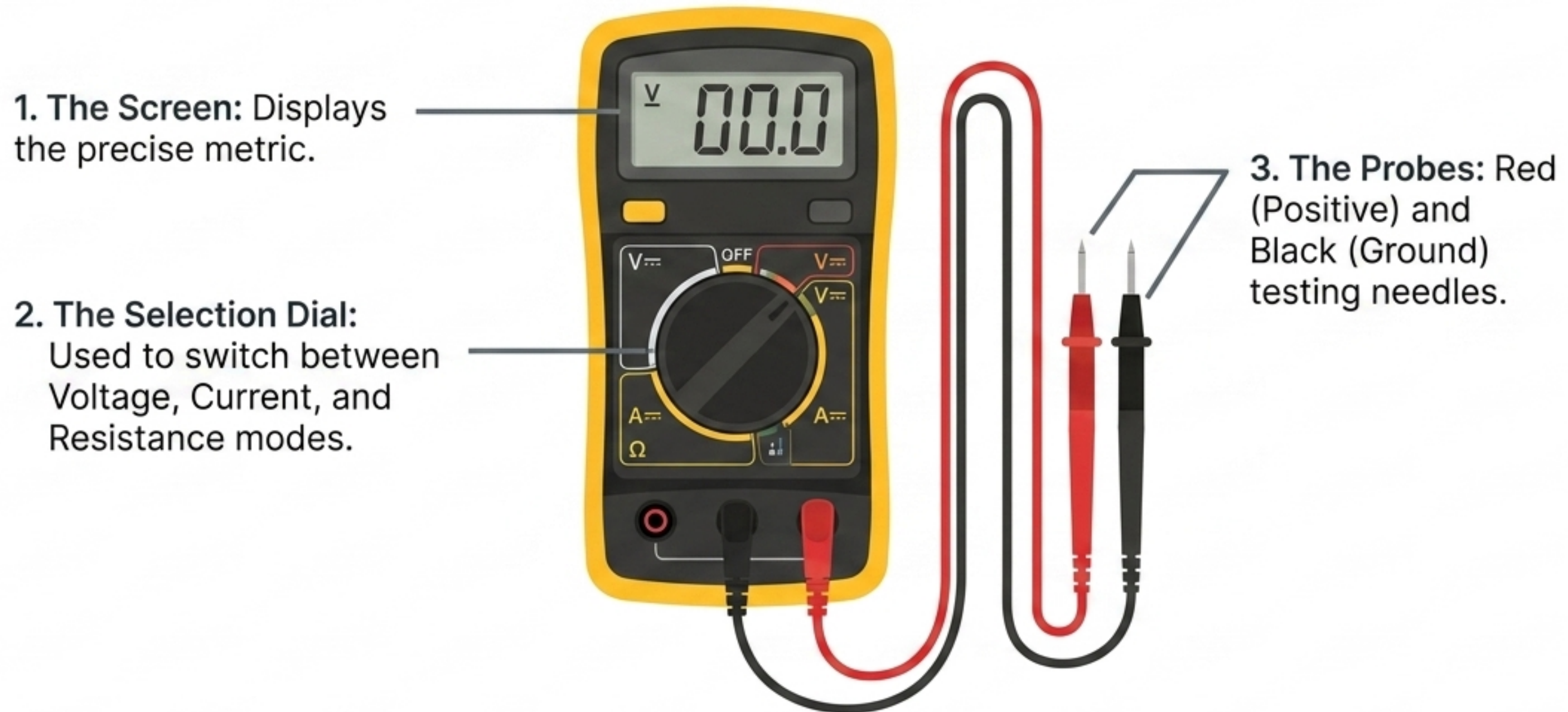
(Relays, Motors, Locks)



Environmental Controls

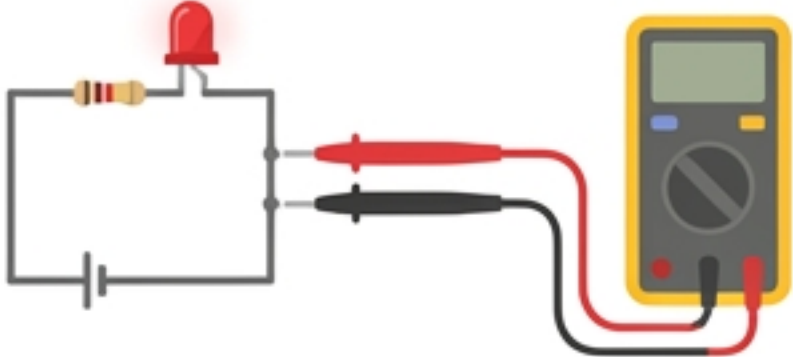
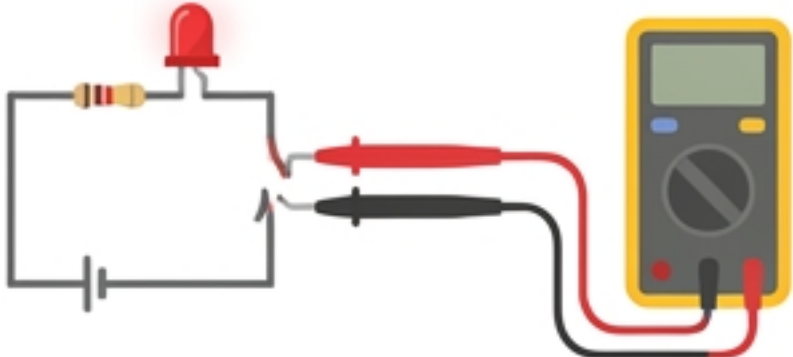
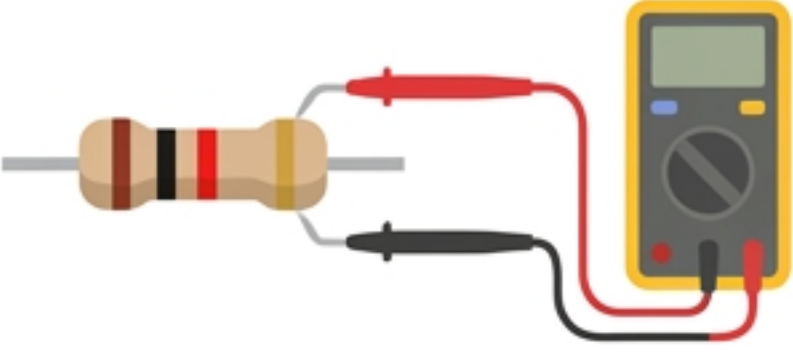
(Agricultural water pumps,
Smart home lighting)

The hardware debugger

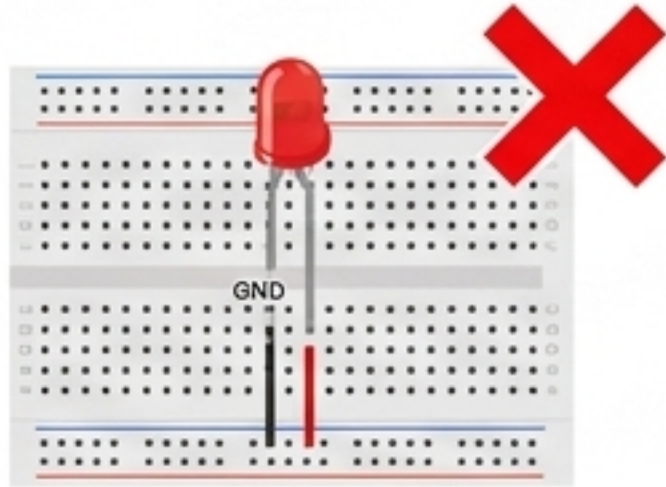


When your IoT prototype fails, the multimeter allows you to 'see' the invisible electricity and find the break in the logic.

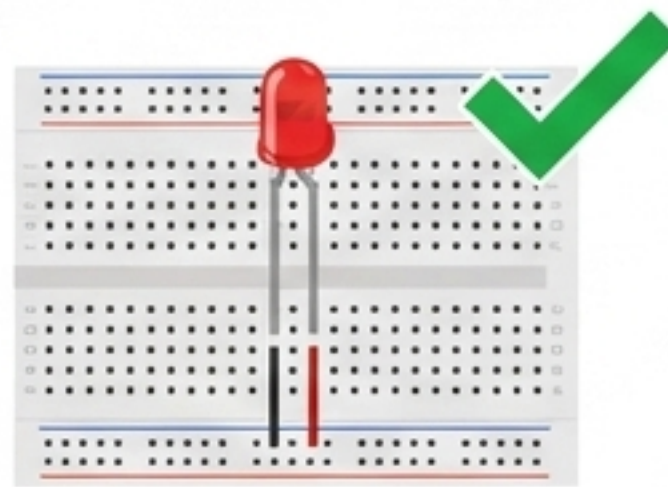
Diagnostic matrix

Measurement	How to Connect	Purpose / Tip
Voltage (V) ●		Checks if power is reaching the destination.
Current (A) ●		Measures actual power consumption.
Resistance (Ω) ●		 CRITICAL: Turn off all power before measuring resistance!

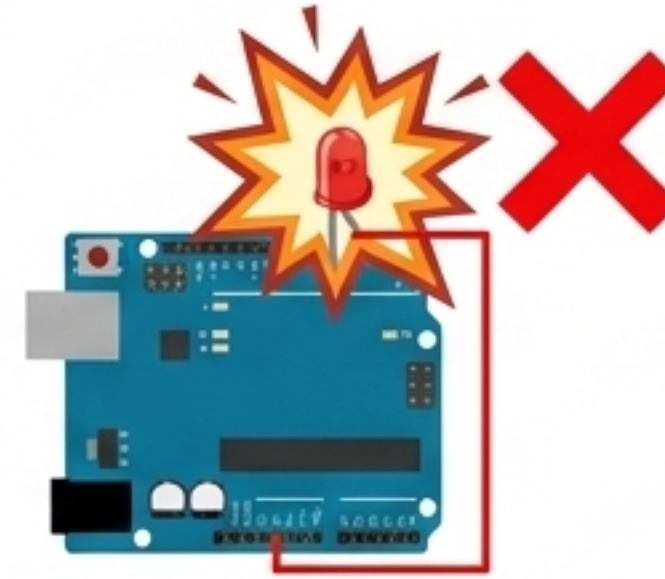
Pre-flight checklist: Avoid these critical errors



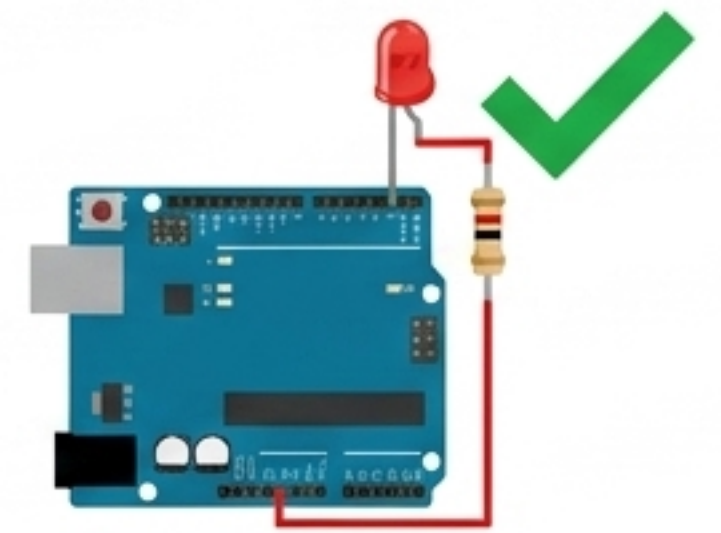
Backward polarity



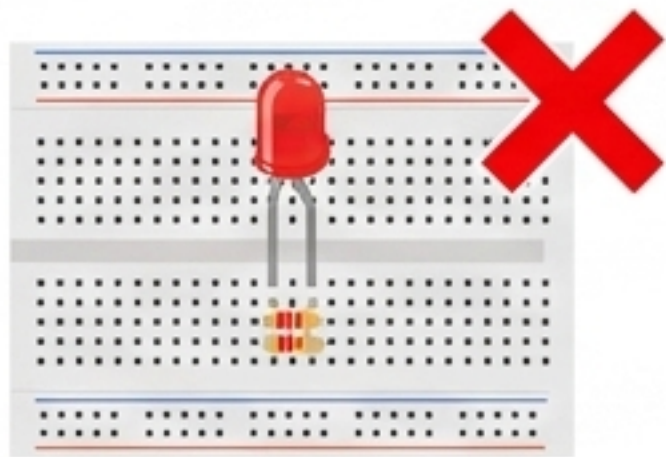
Anode to Power,
Cathode to GND



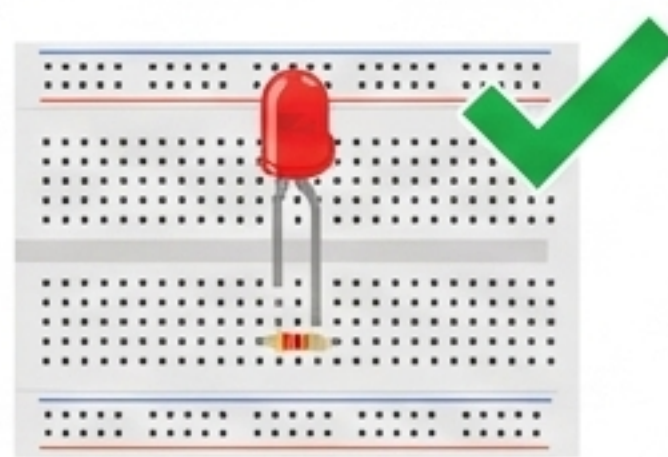
Direct 5V connection



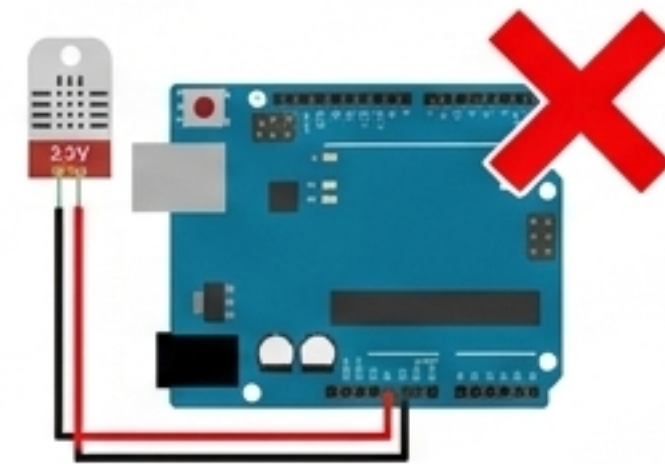
Protected by resistor



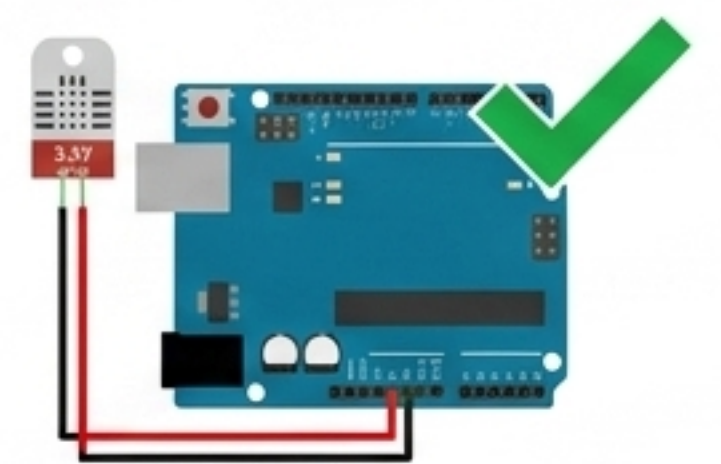
Legs sharing same row
(Short Circuit)



Legs isolated on
separate rows

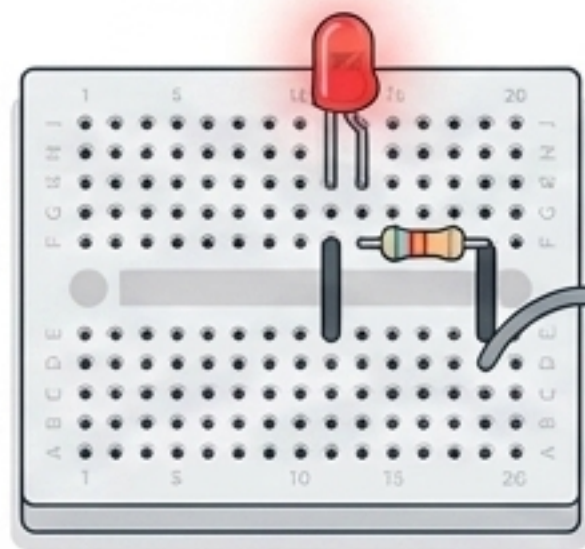


Voltage mismatch

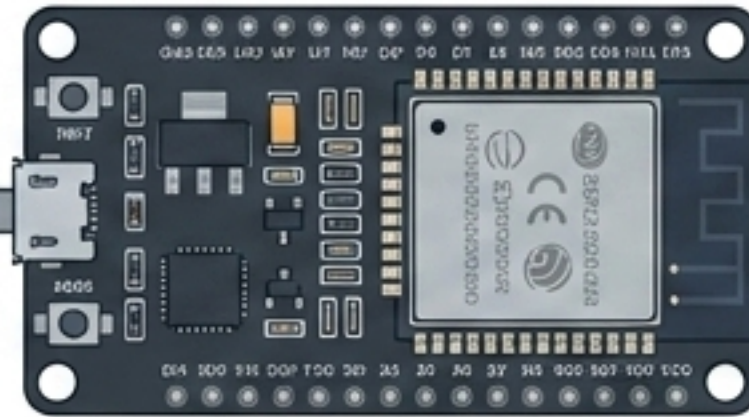


Matched voltage inputs

Scaling up: From the breadboard to the cloud



1. Hardware Output



2. Logic & Control



3. Global IoT Connectivity

You now possess the foundational physical skills. Next, we will introduce code to make these circuits think, react to sensors, and communicate across the internet.